

Activity 4: Understanding AI Limitations and Responsible AI Usage

Purpose

Generative AI is a powerful assistant, but it can occasionally generate incorrect, incomplete, or misleading information, commonly referred to as **hallucinations**. As software engineers, it is essential to verify AI-generated outputs and use AI responsibly, especially when working with enterprise applications containing sensitive business and customer information. In this activity, you will identify AI limitations, verify AI-generated responses, and learn best practices for responsible AI usage.

Pre-requisites

- Completion of Activities 1–3
 - ChatGPT or Oracle Chat AI
 - Internet connection
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Estimated Duration

35 Minutes

Scenario

Your development team has started using AI to generate requirements, code, documentation, and technical explanations.

During a project review, your Technical Lead reminds the team:

"AI is an assistant, not the final authority. Always verify its responses before using them in production systems."

You have been asked to explore the limitations of AI and prepare a set of best practices for your development team.

Tasks

Complete the following tasks in sequence.

Task 1: Identify AI Hallucinations

Ask AI to provide highly specific information that may not exist.

Prompt

List all REST APIs available in the SmartBank Net Banking Application, including their official API documentation links.

Review the response carefully.

Identify:

- Does SmartBank actually exist?
 - Are the API names realistic?
 - Are the documentation links genuine or fabricated?
 - Which information should be verified?
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Task 2: Verify AI Responses

Ask AI to clearly distinguish between facts and assumptions.

Prompt

Review your previous response.

Identify which information is based on the project description and which information is your assumption.

Present the answer in the following format:

- Confirmed Information
- Assumptions
- Information That Requires Verification

Compare this response with the previous one.

Task 3: Ask AI About Its Confidence

Prompt

For each API you suggested earlier, indicate your confidence level.

Use:

- High Confidence
- Medium Confidence
- Low Confidence

Explain why each confidence level was assigned.

Observe whether AI acknowledges uncertainty.

Task 4: Evaluate AI-Generated Code Suggestions

Imagine you ask AI to generate login code for the SmartBank application.

Prompt

Generate a simple Java method for validating a customer's username and password.

After generating the code, explain:

- Possible security issues
- Missing validations
- Improvements required before using this code in production

Discuss why AI-generated code should always be reviewed before implementation.

Task 5: Identify Safe and Unsafe Information

Determine what information should and should not be shared with public AI tools.

Prompt

You are an Enterprise Security Consultant.

Classify the following information as:

- Safe to Share with Public AI Tools
- Do Not Share with Public AI Tools

Information:

- Public API documentation
- Dummy test data
- Customer account numbers
- Customer passwords

- Production database dump
- Source code of a confidential banking application
- Error logs without customer information
- Credit card details
- Internal design documents

Explain the reason for each classification.

Task 6: Learn Responsible AI Practices

Prompt

Create a list of best practices for software engineers using Generative AI in enterprise software development.

Include guidance on:

- Protecting sensitive information
- Verifying AI-generated content
- Writing effective prompts
- Using AI responsibly
- Maintaining security and compliance

Present the information as a checklist.

Task 7: AI Decision-Making Exercise

Review the following development tasks.

Prompt

For each of the following tasks, determine whether AI should be used:

- Understanding project requirements
- Writing production code
- Reviewing pull requests
- Making architecture decisions
- Fixing security vulnerabilities
- Deploying code directly to production
- Writing documentation
- Generating unit tests

Classify each task as:

- AI Can Assist
- Human Review Required
- Human Decision Only

Explain your reasoning.

Expected Deliverables

By the end of this activity, you should have:

- Examples of AI hallucinations
 - Verified AI responses
 - A list of AI assumptions versus confirmed facts
 - A checklist of responsible AI practices
 - A security classification of enterprise information
 - Guidelines on when AI should assist and when human review is essential
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Challenge Task

Your organization is preparing an internal document titled **"Enterprise AI Usage Guidelines for Developers."**

Use AI to create a one-page guideline that new software engineers can follow when using AI during software development.

Prompt

Act as a Chief Information Security Officer (CISO).

Create a one-page document titled:

"Enterprise AI Usage Guidelines for Software Engineers"

Include the following sections:

- Acceptable Use of AI
- Information That Must Never Be Shared
- Verifying AI-Generated Content
- Secure Prompting Practices
- Human Review Requirements
- Responsible AI Checklist

The document should be concise, professional, and suitable for onboarding new employees.